

Chapter 04 Homework — Field extensions

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September 17, 2026

Problem 04.1. Suppose that $[L:K]$ is a prime number. What fields are there intermediate between L and K ?

Solution. Lemma. If $[L:K] = 1$ then $L = K$.

If $[L:K] = 1$, that means you only need one vector of L to span all of L if you are given any coefficients in K . Suppose $\{\ell_1\}$ spans L . Then

$$\forall \ell \in L, \exists k \in K \text{ s.t. } k\ell_1 = \ell.$$

So $k_0\ell_1 = 1$ for some $k_0 \in K$, and surely $k_0 \neq 0$, since that forces $k_0\ell_1 = 0$. Hence $k_0 = \ell_1^{-1}$, so $\ell_1^{-1} \in K$, so $\ell_1 \in K$, since K is a field.

Let $\ell \in L$; show $\ell \in K$. There is k_1 with $k_1\ell_1 = \ell$. Both $k_1, \ell_1 \in K$, so $\ell \in K$.

The exercise. Let $[L:K] = p$ be prime and let M be intermediate, $K \subseteq M \subseteq L$. The tower law gives

$$[L:K] = [L:M][M:K],$$

and the only factorisations of p into positive degrees are

$$\begin{array}{ccc} p & 1 & p \\ p & p & 1 \end{array}$$

that is, $[L:M] = 1$ with $[M:K] = p$, or $[L:M] = p$ with $[M:K] = 1$. By the lemma the first gives $M = L$ and the second gives $M = K$. Both occur:

$$[L:K] = [L:L][L:K] \quad \text{and} \quad [L:K] = [L:K][K:K].$$

Which fields are intermediate? Just L and K .

Problem 04.2. Suppose that $L:K$ and that K_1 and K_2 are two intermediate fields such that $L = K(K_1, K_2)$. Show that $[L:K] \leq [K_1:K][K_2:K]$.

Problem 04.3. Suppose that $K(\alpha):K$ is a finite simple extension. For each β in $K(\alpha)$, let $T_\alpha(\beta) = \alpha\beta$. T_α is a linear mapping of $K(\alpha)$ (considered as a vector space over K) into itself. Show that $\det(xI - T_\alpha)$ is the minimal polynomial of α over K .

Problem 04.4. Show that $x^3 + 3x + 1$ is irreducible in $\mathbb{Q}[x]$. Suppose that α is a root of $x^3 + 3x + 1$ in $\mathbb{C}$. Express α^{-1} and $(1 + \alpha)^{-1}$ as linear combinations, with rational coefficients, of $1, \alpha$ and α^2 .

Solution. Irreducibility. $f(x) = x^3 + 3x + 1 \in \mathbb{Q}[x]$, so the possible rational roots are $\{1, -1\}$.

$$f(-1) = -1 - 3 + 1 \neq 0, \quad f(1) = 1 + 3 + 1 \neq 0.$$

So $f(x)$ has no rational roots and is hence irreducible in $\mathbb{Q}[x]$.

Inverting α . α is a root, so $f(\alpha) = \alpha^3 + 3\alpha + 1 = 0$. Then

$$\alpha^3 + 3\alpha + 1 = 0 \iff -1 = \alpha(\alpha^2 + 3), \quad \text{so} \quad \alpha^{-1} = \underbrace{-1}_{\in \mathbb{Q}} \cdot \alpha^2 \underbrace{-3}_{\in \mathbb{Q}} \cdot 1.$$

Inverting $1 + \alpha$. Set $\beta = 1 + \alpha$, so $\alpha = \beta - 1$. From $\alpha^3 + 3\alpha + 1 = 0$,

$$\begin{aligned} & (\beta - 1)^3 + 3(\beta - 1) + 1 = 0 \\ \iff & (\beta - 1)(\beta^2 - 2\beta + 1) + 3\beta - 3 + 1 = 0 \\ \iff & \beta^3 - 2\beta^2 + \beta - \beta^2 + 2\beta - 1 + 3\beta - 3 + 1 = 0 \\ \iff & \beta^3 - 3\beta^2 + 6\beta - 3 = 0 \\ \iff & 1 = \frac{1}{3}\beta^3 - \beta^2 + 2\beta \\ \iff & 1 = \beta\left(\frac{1}{3}\beta^2 - \beta + 2\right), \end{aligned}$$

so

$$\begin{aligned} \beta^{-1} &= (\alpha + 1)^{-1} = \frac{1}{3}(\alpha + 1)^2 - (\alpha + 1) + 2 \\ &= \frac{1}{3}(\alpha^2 + 2\alpha + 1) - \alpha + 1 \\ &= \underbrace{\frac{1}{3}}_{\in \mathbb{Q}} \alpha^2 \underbrace{- \frac{1}{3}}_{\in \mathbb{Q}} \alpha + \underbrace{\frac{4}{3}}_{\in \mathbb{Q}} \cdot 1 = (\alpha + 1)^{-1}. \end{aligned}$$

Problem 04.5. Suppose that $L(\alpha):L:K$ and that $[L(\alpha):L]$ and $[L:K]$ are relatively prime. Show that the minimal polynomial of α over L has its coefficients in K .

Remark. As printed, 4.5 is false. The tower $K = \mathbb{Q}$, $L = \mathbb{Q}(2^{1/3})$, $\alpha = 2^{1/6}$ has $[L(\alpha):L] = 2$ and $[L:K] = 3$, which are coprime, while the minimal polynomial of α over L is $x^2 - 2^{1/3}$, whose constant term is not rational. The repaired hypothesis is that $[K(\alpha):K]$ and $[L:K]$ are relatively prime. Established in the sitting of 2026-09-15, card 0016.

Problem 04.6. Suppose that $[L:K]$ is a prime number. Show that $L:K$ is simple.

Solution. $[L:K] = p$. L and K are the only intermediate fields (Exercise 4.1).

Note $K(\alpha)$ is by definition the smallest subfield containing $\alpha \in L$ and all $k \in K$.

Pick $\alpha = \ell$ where $\ell \notin K$, which exists since $[L:K] > 1$. Note $K \subsetneq K(\alpha)$.

So $K(\alpha)$ must be L , the only other intermediate subfield between K and L .

Problem 04.7. Show that if $L:K$ is algebraic and K is countable then L is countable. Show that there exist real numbers which are transcendental over the rationals.

Solution. First half: K countable implies $K[x]$ countable. Fix n . A polynomial of degree exactly n is a list of coefficients $k_0, k_1, \dots, k_n$ drawn from K with $k_n \neq 0$, so the polynomials of degree exactly n form a set of $(n+1)$ -tuples drawn from K , and the set of n -tuples drawn from a countable set is countable. The zero polynomial has no degree, so

$$K[x] = \{0\} \cup \bigcup_{n \geq 0} \{f \in K[x] \mid \deg f = n\},$$

a countable union of countable sets together with one extra element. Hence $K[x]$ is countable.

First half: L countable. For each non-zero $f \in K[x]$, denote

$$R_f = \{\ell \in L \mid f(\ell) = 0\}.$$

Each R_f is finite, since a non-zero polynomial of degree n has at most n roots in any field. Further

$$L = \bigcup_{\substack{f \in K[x] \\ f \neq 0}} R_f,$$

because $L:K$ is algebraic, so every $\ell \in L$ is a root of some non-zero $f \in K[x]$ and therefore lies in that R_f . This is a countable union of finite sets — the index set $K[x] \setminus \{0\}$ is countable, and finite sets are countable — and is hence countable.

Second half: a transcendental real exists. Suppose not. Then every real number is algebraic over $\mathbb{Q}$, that is, $\mathbb{R}:\mathbb{Q}$ is algebraic. Since $\mathbb{Q}$ is countable, the first half applies with $K = \mathbb{Q}$ and $L = \mathbb{R}$, and gives that $\mathbb{R}$ is countable. But $\mathbb{R}$ is not countable. So the supposition is false, and there is a real number transcendental over $\mathbb{Q}$.

Problem 04.8. Suppose that $L:K$ is an extension, that α is an element of L which is transcendental over K , and that f is a non-constant element of $K[x]$. Show that $f(\alpha)$ is transcendental over K . Show that, if β is an element of L which satisfies $f(\beta) = \alpha$, then β is transcendental over K .

Solution. First half. Suppose there were a non-zero $g \in K[x]$ with $g(f(\alpha)) = 0$. Then α is algebraic over K , since $g \circ f \in K[x]$ and is non-zero. But α is transcendental over K , a contradiction. Hence $f(\alpha)$ is transcendental over K .

Second half. Suppose β were algebraic over K . Then $[K(\beta):K]$ is finite, and every element of a finite extension of K is algebraic over K . Now $f = c_n x^n + \cdots + c_1 x + c_0$ with every $c_i \in K$, and $K(\beta)$ is a field containing K and β , so it is closed under multiplication — each $c_i \beta^i \in K(\beta)$ — and under addition, so the sum is too. Hence

$$\alpha = f(\beta) \in K(\beta),$$

and so α is algebraic over K . But α is transcendental over K , a contradiction. Hence β is transcendental over K .

Problem 04.9. Suppose that a and b are complex numbers which are transcendental over $\mathbb{Q}$. Is ab transcendental over $\mathbb{Q}$?

Solution. No. The product of two transcendental numbers need not be transcendental.

Take $a = \pi$ and $b = 1/\pi$. Both are transcendental over $\mathbb{Q}$: π by Lindemann's theorem, and $1/\pi$ by the reversal argument quoted above. Their product is $ab = 1$, which is a root of $x - 1$, a non-zero polynomial in $\mathbb{Q}[x]$. So ab is algebraic over $\mathbb{Q}$.

The same thing in general: in any extension $L:K$, both a and b can lie outside K while ab lies inside it.

Problem 04.10. Suppose that $K(\alpha, \beta):K$ is an extension, that α is algebraic over K , but not in K , and that β is transcendental over K . Show that $K(\alpha, \beta):K$ is not simple.

Solution. Suppose, for a contradiction, that the extension is simple: there is a single $\gamma \in K(\alpha, \beta)$ with $K(\gamma) = K(\alpha, \beta)$. Over K , the element γ is either algebraic or transcendental, and there is no third case, so both must be ruled out.

Case 1: γ algebraic over K . Then $[K(\gamma):K]$ is finite, and every element of a finite extension of K is algebraic over K . Now $\beta \in K(\gamma)$, because $K(\gamma) = K(\alpha, \beta)$. So β is algebraic over K , contradicting the hypothesis that β is transcendental over K .

Case 2: γ transcendental over K . Consider the tower

$$K \subseteq K(\alpha) \subseteq K(\alpha, \gamma).$$

First, $[K(\alpha):K]$ is finite, because α is algebraic over K and so has a minimal polynomial over K of which it is a root.

Second, $[K(\alpha, \gamma):K(\alpha)]$ is finite. Since $K(\gamma) = K(\alpha, \beta)$ we have $\alpha \in K(\gamma)$, so

$$\alpha = \frac{f(\gamma)}{g(\gamma)} \quad \text{for some } f, g \in K[x] \text{ with } g(\gamma) \neq 0,$$

and hence $f(\gamma) - \alpha g(\gamma) = 0$. The polynomial $f(x) - \alpha g(x)$ lies in $K(\alpha)[x]$ and is not the zero polynomial: if it were, then $f_i = \alpha g_i$ for every i , and picking one index i with $g_i \neq 0$ gives $\alpha = f_i/g_i \in K$, contradicting $\alpha \notin K$. So γ is a root of a non-zero polynomial over $K(\alpha)$, i.e. γ is algebraic over $K(\alpha)$, and therefore has a minimal polynomial over $K(\alpha)$, making $[K(\alpha, \gamma):K(\alpha)]$ finite.

By the tower law,

$$[K(\alpha, \gamma):K] = [K(\alpha, \gamma):K(\alpha)] \cdot [K(\alpha):K],$$

a product of two finite numbers, so $[K(\alpha, \gamma):K]$ is finite. Then $[K(\gamma):K]$ is finite too, since $K \subseteq K(\gamma) \subseteq K(\alpha, \gamma)$ and the tower law makes $[K(\gamma):K]$ a factor of a finite number. A finite extension has every element algebraic over the base field, so γ is algebraic over K — contradicting the assumption of this case, that γ is transcendental over K .

Both cases are impossible, so no such γ exists and $K(\alpha, \beta):K$ is not simple.

Problem 04.11. Show that the condition that $L:K$ is algebraic cannot be dropped from Theorem 4.8.